

Chinese Policy Text Analysis

A Practice Based on “教育强国建设规划纲要（2024–2035 年）”

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《教育强国建设规划纲要（2024—2035年）》主要内容如下。

为深入贯彻党的二十大关于加快建设教育强国的战略部署，制定本规划纲要。

一、总体要求

坚持以习近平新时代中国特色社会主义思想为指导，深入贯彻党的二十大和二十届二中、三中全会精神，全面贯彻习近平总书记关于教育的重要论述，深刻领悟“两个确立”的决定性意义，坚决做到“两个维护”，坚持教育优先发展，全面贯彻党的教育方针，坚定不移走中国特色社会主义教育发展道路，坚持社会主义办学方向，全面把握教育的政治属性、人民属性、战略属性，落实立德树人根本任务，为党育人、为国育才，全面服务中国式现代化建设，扎根中国大地办教育，加快建设高质量教育体系，培养德智体美劳全面发展的社会主义建设者和接班人，加快建设具有强大思政引领力、人才竞争力、科技支撑力、民生保障力、社会协同力、国际影响力的中国特色社会主义教育强国，为建设社会主义现代化强国、全面推进中华民族伟大复兴提供有力支撑。

工作中要做到：坚持党对教育事业的全面领导，突出促进公平、提高质量，强化战略引领、支撑发展，深化改革创新、协同融合，坚持自主自信、胸怀天下。正确处理支撑国家战略和满足民生需求、知识学习和全面发展、培养人才和满足社会需要、规范有序和激发活力、扎根中国大地和借鉴国际经验的关系，全面构建固本铸魂的思想政治教育体系、公平优质的基础教育体系、自强卓越的高等教育体系、产教融合的职业教育体系、泛在可及的终身教育体系、创新牵引的科技支撑体系、素质精良的教师队伍体系、开放互鉴的国际合作体系，实现由大到强的系统跃升。

主要目标是：到2027年，教育强国建设取得重要阶段性成效。各级教育普及水平持续巩固提升，高质量教育体系初步形成，人民群众教育获得感明显提升，人才自主培养质量全面提高，拔尖创新人才不断涌现，关键领域改革取得实质性进展，教育布局结构与经济社会和人口高质量发展需求更加契合，具有全球影响力的重要教育中心建设迈上新台阶。到2035年，建成教育强国。党对教育事业全面领导的制度体系和工作机制系统完备，高质量教育体系全面建成，基础教育普及水平和质量稳居世界前列，学习型社会全面形成，人民群众教育满意度显著跃升，教育服务国家战略能力显著跃升，教育现代化总体实现。

Task and Data

- Text file: Text_File.txt
- Source text: “教育强国建设规划纲要”
- Text size: 9310 Chinese characters, 2690 tokens, 1038 unique tokens
- Output types: visualizations + an interpretive report
- **Visual outputs:** word cloud, top-word chart, keyword chart, co-occurrence heatmap, bigram network, section heatmap, theme heatmap, action-verb chart, and Zipf curve
- **Textual outputs:** key sentences, section summaries, policy-language features, and theme interpretation

Workflow

- ➊ Read the raw text and normalize line breaks and whitespace
- ➋ Use [jieba](#) for Chinese tokenization and register domain phrases
- ➌ Build a [stopword list](#) to remove function words and low-information tokens
- ➍ Count [word frequencies](#) and generate the word cloud and top-word chart
- ➎ Extract keywords with [TF-IDF and TextRank](#)
- ➏ Build a sliding-window co-occurrence matrix for word relationships
- ➐ Extract [bigrams](#) and filter them with frequency
- ➑ [Split the document by major sections](#) to compare thematic focus
- ➒ Add interpretive layers such as key sentences, action verbs, and theme dimensions

Tokenization using Jieba

```
import jieba
text = "南京市长江大桥欢迎您"
jieba.lcut(text) # 精确模式 (默认)
jieba.lcut(text, cut_all=True) # 所有可能的组合
jieba.cut_for_search(text) # 标准答案 + 精确分词 + 子词补充
```

Results for the above three lines of codes

['南京市', '长江大桥', '欢迎您']

['南京', '南京市', '南京市长', '京市', '市长', '长江', '长江大桥', '江大桥', '大桥', '欢迎', '欢迎您']

['南京', '京市', '南京市', '长江', '大桥', '江大桥', '长江大桥', '欢迎', '欢迎您']

Update the Database

```
import jieba
text = "南京市长江大桥欢迎您"
jieba.add_word("江大桥", freq=10000000000)
print(jieba.lcut(text)) # 精确模式 (默认)
```

Results for the above code is

```
['南京市长', '江大桥', '欢迎您']
```

Conclusion: Jieba relies on a **built-in dictionary of frequently occurring words and phrases** to produce reasonable segmentations. When working in a **different domain**, this dictionary often needs to be **updated** or extended (e.g., by adding domain-specific terms or adjusting frequencies) to achieve accurate results.

First Task: Chinese Tokenization and Filtering

```
StopWords = {"一个", "一些", "一种", "一批", "不是", "以及", "各", "各类", "各项", "各级",  
            "各方面", "同时", "和", "总体", "工作", "方面", "进一步", "通过", "重要"}  
Custom_Words = ["教育强国", "立德树人", "思想政治教育", "基础教育", "高等教育", "职业教育",  
               "教师队伍", "人才培养", "创新人才", "数字教育", "国际合作", "课程教材"]
```

- We define our stopwords based on the context.
- Chinese text cannot be split directly by spaces, so tokenization is necessary.
- Part-of-speech tags help separate content words, action verbs, and noisy tokens.
- Custom phrases preserve **full concepts** such as “教育强国”, “职业教育”, and “教师队伍”.

Chinese Tokenization and Filtering

```
TEXT_PATH = Path("Text_Data_Application/Text_File.txt")
text = TEXT_PATH.read_text(encoding="utf-8")
text = text.replace("\n", "") # 去掉换行符
text = text.replace("\u3000", "") # 去掉空格
tokens = []
for word in jieba.lcut(text): # Text Segmentation
    word = word.strip() # 去掉字符串两端的字符
    if len(word) < 2: # 跳过长度 < 2 的词
        continue
    if word in StopWords: # 跳过stopwords
        continue
    if not re.fullmatch(r"[\u4e00-\u9fffA-Za-z0-9]+", word):
        # 保留中文\u4e00-\u9fff 英文字母A-Z a-z, 数字0-9
        continue
    tokens.append(word)
```

- Remove punctuation and construct the token set.
- **Question:** What is our next step?

Analyze frequencies of words

```
word_freq = Counter(tokens)
top_words = word_freq.most_common(20)
print("Top 10 words:", top_words[:10])
```

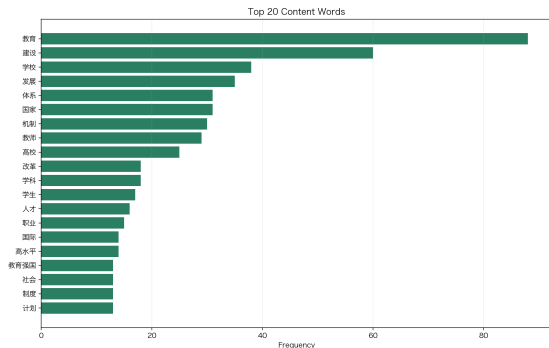
The result is

```
[('教育', 98), ('建设', 64), ('加强', 47), ('学校', 39),
 ('体系', 38), ('完善', 37), ('发展', 35), ('国家', 35),
 ('机制', 30), ('教师', 29)]
```

We run the word cloud.

```
wc = WordCloud(font_path=FONT_PATH,width=1400,height=900,background_color="white",colormap="viridis").
    generate_from_frequencies(word_freq)
plt.imshow(wc)
plt.show()
```

Word Cloud and High-Frequency Words



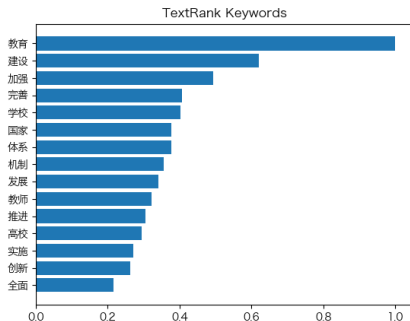
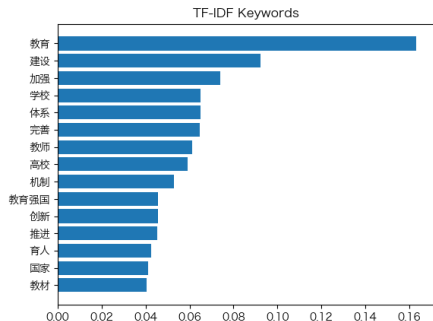
- The most frequent words cluster around terms such as “教育”, “建设”, “学校”, “体系”, and “教师”.
- This indicates a very concentrated policy focus around education-system construction.

Keyword Extraction: TF-IDF and TextRank

```
tfidf_keywords = jieba.analyse.extract_tags(text, topK=15, withWeight=True)
textrank_keywords = jieba.analyse.textrank(text, topK=15, withWeight=True)
def plot_keywords(keywords, title):
    words = [w for w, _ in keywords]
    weights = [w for _, w in keywords]
    plt.barh(words, weights)
    plt.gca().invert_yaxis()
    plt.show()
plot_keywords(tfidf_keywords, "TF-IDF Keywords")
plot_keywords(textrank_keywords, "TextRank Keywords")
```

- We can use two different metrics to extract word importance, including TF-IDF and TextRank.

Frame Title



- TF-IDF emphasizes distinctiveness across paragraphs.
- TextRank emphasizes contextual connectivity inside the document.
- Both methods point to the same core areas: “教育强国”, system building, teachers and talent, and vocational education.

Second Task: Word Co-occurrence

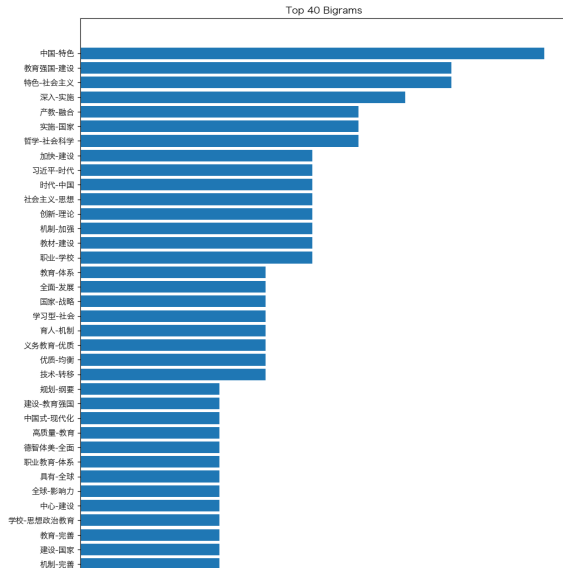
```
bigrams = []
for i in range(len(tokens) - 1):
    left = tokens[i]
    right = tokens[i + 1]
    if left != right:
        bigrams.append((left, right))
bigram_freq = Counter(bigrams)
top_bigrams = bigram_freq.most_common(40)
```

- Here, “relatedness” is defined as **co-occurrence** inside a sliding window.
- A larger window makes the network denser, while a smaller window makes relationships more local.

Second Task: Word Co-occurrence

```
labels = [f"{w1}-{w2}" for (w1,w2),_ in
          top_bigrams]
values = [v for _,v in top_bigrams]
plt.barh(labels[:40], values[:40])
plt.title("Top 40 Bigrams")
plt.show()
```

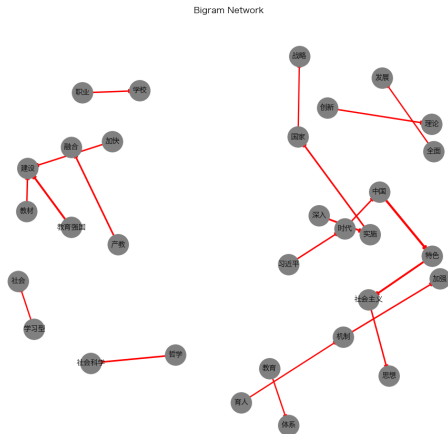
- **Question:** What message can we take from this result?



Network View of Word Co-occurrence

```
G = nx.DiGraph()
for (left, right), weight in top_bigrams[:20]:
    G.add_edge(left, right, weight=weight)
fig, ax = plt.subplots(figsize=(10, 9))
pos = nx.spring_layout(G, k=1.0, seed=42)
edge_widths = [1 + G[u][v]["weight"] * 0.2 for u, v in G.edges()]
nx.draw_networkx_nodes(G, pos, node_color="grey", node_size=800, ax=ax)
nx.draw_networkx_edges(G, pos, width=edge_widths, edge_color="red", arrows=True, ax=ax)
nx.draw_networkx_labels(G, pos, font_family="Hiragino Sans GB", font_size=10, ax=ax)
ax.set_title("Bigram Network")
ax.axis("off")
plt.tight_layout()
plt.show()
```

Network View of Word Co-occurrence



- Typical pairs include “教育强国 → 建设”, “义务教育 → 优质”, and “学校 → 思想政治教育”.

Section View and Frequency Distribution

```
section_matches = list(re.finditer(r"(?m)^(一|二|三|四|五|六|七|八|九|十|+|.+)$", text))
sections = []
for i, match in enumerate(section_matches):
    start = match.end()
    end = section_matches[i + 1].start() if i + 1 < len(section_matches) else len(text)
    sections.append((match.group(1).strip(), text[start:end].strip()))

top_terms = [word for word, _ in top_words[:8]]
section_rows = []
for section_title, section_body in sections:
    section_tokens = [w for w in jieba.lcut(section_body) if len(w) >= 2 and w not in StopWords]
    section_counts = Counter(section_tokens)
    total = sum(section_counts.values()) or 1
    row = {"section": section_title}
    for term in top_terms:
        row[term] = section_counts.get(term, 0) / total * 1000
    section_rows.append(row)
```

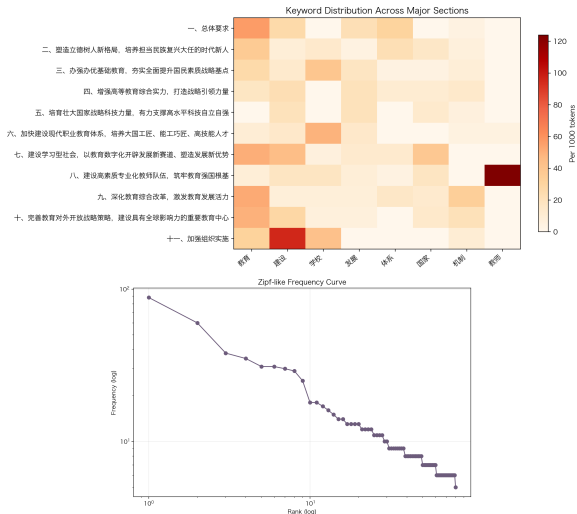
- We split the document by **major section headings**.
- Then we compare how often important words appear in each section.

Section View and Frequency Distribution

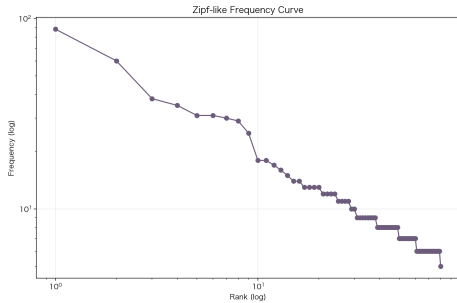
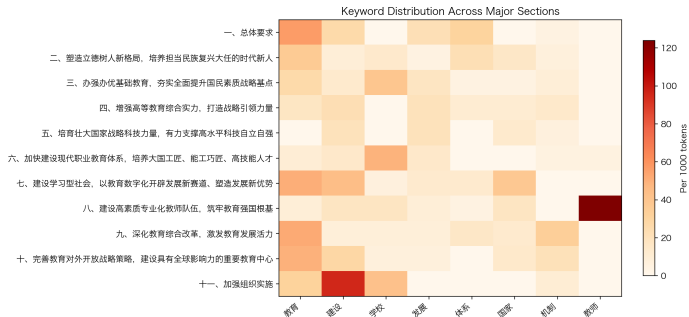
```
section_term_matrix = pd.DataFrame(  
    section_rows).set_index("section")  
plt.imshow(section_term_matrix.values, cmap  
    ="OrRd", aspect="auto")  
plt.xticks(range(len(top_terms)), top_terms  
    , rotation=40)  
plt.yticks(range(len(section_term_matrix.  
    index)), section_term_matrix.index)  
plt.show()
```

```
zipf_counts = np.array([count for _, count  
    in word_freq.most_common(80)])  
zipf_ranks = np.arange(1, len(zipf_counts)  
    + 1)  
plt.loglog(zipf_ranks, zipf_counts, marker=  
    "o")  
plt.show()
```

- **Question:** What can we learn from the section view?



Section View and Frequency Distribution



- The section heatmap shows that different major sections emphasize different keyword clusters.
- The Zipf curve shows that a small number of high-frequency words carry much of the semantic weight.

Theme Dimensions and Policy Action Verbs

```
theme_rows = []
for section_title, section_body in sections:
    section_tokens = [w for w in jieba.lcut(section_body) if len(w) >= 2 and w not in StopWords]
    section_counts = Counter(section_tokens)
    row = {"section": section_title}
    for theme_name, theme_words in THEME_KEYWORDS.items():
        row[theme_name] = sum(section_counts.get(w, 0) for w in theme_words)
    theme_rows.append(row)

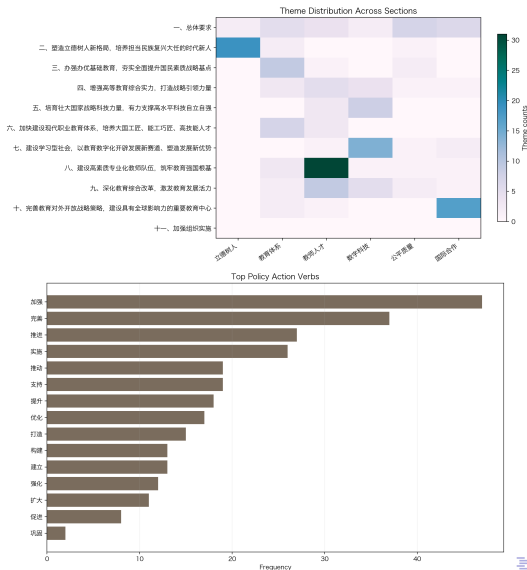
action_counts = Counter()
for item in pseg.cut(text):
    if item.word in ACTION_VERBS and item.flag.startswith(("v", "vn")):
        action_counts[item.word] += 1
```

- We manually define several **theme dimensions**.
- We also count **action verbs** to capture policy style.

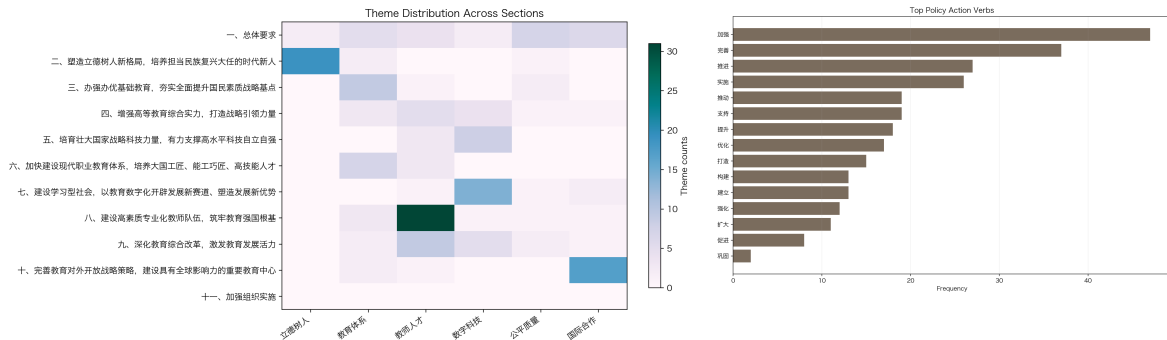
Theme Dimensions and Policy Action Verbs

```
theme_matrix = pd.DataFrame(theme_rows).  
    set_index("section")  
plt.imshow(theme_matrix.values, cmap="PuBuGn", aspect="auto")  
plt.show()  
  
top_actions = action_counts.most_common(15)  
labels = [w for w, _ in top_actions]  
values = [c for _, c in top_actions]  
plt.barh(labels[::-1], values[::-1])  
plt.show()
```

- **Question:** Why are action verbs useful in policy text?



Theme Dimensions and Policy Action Verbs



- The theme heatmap compresses the full document into six major dimensions.
- The action-verb chart shows that the text is not merely descriptive; it is strongly action-oriented and governance-oriented.

How to Explain the Textual Analysis

```
top_action_words = [word for word, _ in action_counts.most_common(6)]
dominant_themes = theme_matrix.sum().sort_values(ascending=False)

print(top_action_words)
print(dominant_themes)
```

- We combine **theme strength**, **section focus**, and **action verbs**.
- Then we interpret the document as a structured policy plan rather than a loose collection of ideas.

How to Explain the Textual Analysis

- At the sentence level, the most important statements concentrate around overall goals, education reform, and support for science and talent.
- At the section level, Part II emphasizes “立德树人”, Part VIII emphasizes “教师队伍”, and Part X emphasizes “国际合作”.
- At the language level, frequent action verbs such as “加强”, “完善”, “推进”, and “实施” show an implementation-oriented policy style.
- At the interpretation level, this is not a single-issue reform text but a macro planning document structured as “overall goal + subsystems + execution levers”.

Key Takeaways

- The core theme is the systematic construction of “教育强国” through a high-quality education system.
- Major levers include “立德树人”, school-system coordination, vocational education, teacher development, talent cultivation, digital education, and international cooperation.
- Results from frequency, keywords, co-occurrence, bigrams, action verbs, and theme dimensions all support a “central theme + subtopics + policy levers” structure.
- This case is well suited for teaching how text mining turns a long policy document into structured evidence and then back into interpretation.